CSCI E-102 Econometrics and Causal Inference with R

Harvard Extension School

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Spring 2026
Lecture 1

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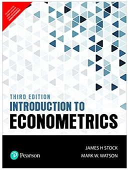
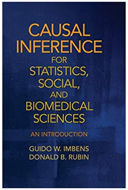
CSCI E-102 Course Information: Textbooks/Readings

Required text:
James H. Stock and Mark W. Watson,
Introduction to Econometrics, 3rd ed., 2017
ISBN: 978-9-352-86350-1
Alternatively, you can use the 4th edition of the book.



Optional reading:
Guido W. Imbens and Donald B. Rubin,
Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction, 1st ed., 2015
ISBN: 978-0-521-88588-1
Electronic copy of the book is available via Harvard Library.



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CSCI E-102 Course Information: Prerequisites

Calculus

Calculus equivalent to MATH E-15:

- Geometric series
- Limits
- Derivatives
- Chain Rule
- Optimization
- Integrals
- Fundamental Theorem of Calculus
- Taylor series etc.

Introductory probability and statistics:

- Random variables
- Probability distributions
- Expectations
- Joint distributions
- Conditional expectations etc.

CSCI E-102 Course Information: Prerequisites

R

Prior programming experience, preferably in R, is helpful but not required.

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CSCI E-102 Course Information: Grading

Assignments, Quizzes, Exams

$$\begin{aligned} \text{Grade} &= 0.25 \cdot \text{Homework (weekly, starting February 8)} \\ &+ 0.20 \cdot \text{Quizzes (weekly, starting February 3)} \\ &+ 0.25 \cdot \text{Midterm (due March 22, 11:59 pm ET)} \\ &+ 0.30 \cdot \text{Final (due May 12, 11:59 pm ET)} \end{aligned}$$

Contents

CSCI E-102 Course Information: Dates of Interest

- The course starts, January 27
- Last day to change the credit status, February 2
- Course drop deadline for full-tuition refund, February 2
- Quiz 1 is due, February 3
- Assignment 1 is due, February 8
- Course drop deadline for half-tuition refund, February 9
- **Midterm Exam** is due, March 22, 11:59 pm (Eastern Time)
- Withdrawal deadline, April 17
- **Final Exam** is due, May 12, 11:59 pm (Eastern Time)

Contents

Supervised & Unsupervised ML

- Supervised Learning
  - Linear Regression
  - Logistic Regression
  - Linear Discriminant Analysis (LDA)
  - Quadratic Discriminant Analysis (QDA)
  - Naive Bayes
  - K-nearest Neighbors (KNN)
  - Random Tree / Forest
  - Support Vector Machine (SVM)
  - Artificial Neural Networks (NN)
- Unsupervised Learning
  - K-means Clustering
  - Hierarchical Clustering
  - Mixture Model
  - Expectation–maximization (EM) Algorithm
  - Method of Moments
  - Artificial Neural Networks (NN)

Deep Reinforcement Learning

- k-armed Bandit Problem: ε-greedy Selection, Upper-Confidence-Bound Action Selection, and Gradient Bandit Algorithms
- Dynamic Programming: Policy Evaluation, Policy Improvement, and Policy Iteration
- Monte Carlo (MC) Methods: MC Estimation and Control
- Off-policy Learning: Importance Sampling
- One-step Temporal-Difference (TD) Learning
- SARSA
- Q-learning and Double Q-learning
- n-step Methods: n-step TD Prediction and n-step SARSA
- Approximate Solution Methods: Semi-gradient 1-step TD for Prediction, Semi-gradient 1-step SARSA for Control, Double Q-Learning with Approximation, Deep Q-Network, TD(λ), and SARSA(λ)
- Policy Gradient Methods: REINFORCE, REINFORCE with Baseline, and Actor-Critic Methods

Contents

Econometric Models

- Multivariate Linear Regression
- Heteroscedasticity and Weighted Least Squares (WLS)
- Dummy Variables and Interactions
- Difference in Differences (DD)
- Logistic Regression
- Probit Model
- Censored Regression Models
- Exact Matching
- Propensity Score Matching (PSM)
- Regression Discontinuity Design (RDD)
- Fuzzy Regression Discontinuity (FRD)
- Synthetic Control
- Instrumental Variables (IV)
- Two-stage Least Squares (2SLS)

Contents

Definition of Random Variable

Def.

A *random variable* (often abbreviated as r.v.) $X(\omega)$ is a function from a sample space Ω to $\mathbb{R}$.

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Examples:

1. Toss a coin 100 times and let X = number of heads observed.
2. Roll two dice and let X = sum of the outcomes.
3. For $A \subseteq \Omega$, let
$$I_A(\omega) = \begin{cases} 1, & \text{if } \omega \in A, \\ 0, & \text{if } \omega \notin A. \end{cases}$$
4. Pick a real number uniformly at random from $[0, 1]$ and call it X.
5. Let X be the length of a randomly selected telephone call.

Cumulative Distribution Function (cdf)

Def.
The cumulative distribution function (cdf) of a random variable X is defined as follows:

$$F_X(x) = P(X \leq x) \text{ for all } x \in \mathbb{R}.$$

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Example:
Let X be a r.v. that takes values 1, 2, and 3 with probabilities 0.7, 0.2, and 0.1, respectively.
Then

$$F_X(x) = \begin{cases} 0, & \text{if } x < 1, \\ 0.7, & \text{if } 1 \leq x < 2, \\ 0.9, & \text{if } 2 \leq x < 3, \\ 1, & \text{if } x \geq 3. \end{cases}$$

Definition of a Discrete Random Variable

Def.

A random variable X is called *discrete* if it can take at most a countable number of values.

Definition of a Discrete Random Variable

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A random variable X is called *discrete* if it can take at most a countable number of values.

Examples:

1. Let X be a r.v. that takes values 0, 1, 2, and 3 with $P(X = 0) = 0.25$, $P(X = 1) = 0.125$, $P(X = 2) = 0.125$, and $P(X = 3) = 0.5$.
2. Fix $\lambda > 0$. Let X be a r.v. that takes values $0, 1, 2, \ldots$ with $P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$ for $k \in \{0, 1, 2, \ldots\}$
3. Three identical fair coins are thrown simultaneously until all three show the same face. Let X be the number of times they are thrown.

Probability Mass Function (pmf)

Def.

Let X be a discrete random variable that takes values $x_1, x_2, x_3, \ldots$
The probability mass function (pmf) of X is defined as follows:

$$p_X(x_i) = P(X = x_i) \text{ for all } i = 1, 2, 3, \ldots$$

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Example:
Suppose a die is loaded such that 1, 2, 3, 4, and 5 are equally likely while 6 is twice more likely to appear than any other outcomes. The die is rolled once and the outcome is denoted by X. Then

$$p_X(x) = \begin{cases} \frac{1}{7}, & \text{if } x \in \{1, 2, 3, 4, 5\}, \\ \frac{2}{7}, & \text{if } x = 6. \end{cases}$$

Definition of a Continuous Random Variable

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A random variable X is called *continuous* if its cdf $F_X(x)$ is a continuous function of x.

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Examples:

1. Pick a real number uniformly at random from $[0, 1]$ and call it X. In this case, we will write
$$X \sim \text{Unif}[0, 1].$$
2. Let X be the length of a randomly selected telephone call.
3. Assume $X \sim \text{Unif}[0, 1]$ and let $Y = X^2$.

Probability Density Function (pdf)

<u>Def.</u>
Let X be a continuous random variable.
The probability density function (pdf) of X is the function $f_X(x)$ that satisfies

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt \text{ for all } x \in \mathbb{R}.$$

Probability Density Function (pdf)

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Example:

Let $X \sim$ Unif$[0, 1]$ then

$$F_X(x) = \begin{cases} 0, & \text{if } x < 0, \\ x, & \text{if } 0 \leq x \leq 1, \\ 1, & \text{if } x > 1, \end{cases}$$

and

$$f_X(x) = \begin{cases} 0, & \text{if } x < 0, \\ 1, & \text{if } 0 \leq x \leq 1, \\ 0, & \text{if } x > 1. \end{cases}$$

Expected Value of a Discrete Random Variable

Def.

Let X be a discete random variable which takes values $x_1, x_2, x_3, \ldots$

The *expected value* (or *expectation* or *mean*) of X, denoted by $E[X]$, is defined as follows:

$$E[X] = \sum_i x_i P(X = x_i).$$

If X takes infinitely many values and the series is divergent, the expected value is not defined.

Expected Value of a Discrete Random Variable

Examples:

1. Let $X \sim$ Bernoulli(p) with $p \in [0, 1]$ then
$$E[X] = p.$$
2. Let $X \sim$ Binomial(n, p) with $n \in \{1, 2, 3, \dots\}$ and $p \in [0, 1]$ then
$$E[X] = np.$$
3. Let $X \sim$ Geometric(p) with $p \in (0, 1]$ then
$$E[X] = \frac{1}{p}.$$
4. Let X take $1, 2, 3, \dots$ values with $P(X = k) = \frac{6}{\pi^2} \frac{1}{k^2}$ for all $k \in \{1, 2, 3, \dots\}$ then
$$E[X] \ \text{ is not defined.}$$

Expected Value of a Continuous Random Variable

Def.

Let X be a continuous random variable with probability density function (pdf) $f(x)$. The *expected value* (or *expectation* or *mean*) of X, denoted by $E[X]$, is defined as follows:

$$E[X] = \int_{-\infty}^{+\infty} x f(x) dx.$$

If the integral is divergent, the expected value is not defined.

Expected Value of a Continuous Random Variable

Examples:

1. Let $X \sim \text{Uniform}[a, b]$ with $a < b$, that is,

$$f_X(x) = \begin{cases} 0, & \text{if } x < a, \\ \frac{1}{b-a}, & \text{if } a \leq x \leq b, \\ 0, & \text{if } x > b, \end{cases}$$

   then

$$E[X] = \frac{a+b}{2}.$$

2. Let $X \sim N(0, 1^2)$, i.e. be *standard normal*, that is,

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for all } x \in \mathbb{R},$$

   then

$$E[X] = 0.$$

Variance and Standard Deviation

Def.

Let X be a random variable. Then, provided the expectations exist,

- Variance of X is defined as

$$\mathrm{Var}(X) = \mathrm{E}\left[(X - \mathrm{E}[X])^2\right].$$

- Note that $\mathrm{Var}(X) \geq 0$. Standard deviation of X is defined as

$$\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}.$$

Properties of Variance

Let X be a random variable for which $\mathrm{Var}(X)$ exists and $a, b \in \mathbb{R}$ be some constants, then

1. $\mathrm{Var}(X) \geq 0$
2. $\mathrm{Var}(a) = 0$
3. $\mathrm{Var}(X) = \mathrm{E}[X^2] - (\mathrm{E}[X])^2$
4. $\mathrm{Var}(a + bX) = b^2 \mathrm{Var}(X)$
5. $\mathrm{SD}(a + bX) = |b| \mathrm{SD}(X)$

Variance of a Discrete Random Variable

Examples:

1. Let $X \sim$ Bernoulli(p) with $p \in [0, 1]$ then
$$\mathrm{E}[X] = p \text{ and } \mathrm{Var}(X) = p(1-p).$$
2. Let $X \sim$ Binomial(n, p) with $n \in \{1, 2, 3, \ldots\}$ and $p \in [0, 1]$ then
$$\mathrm{E}[X] = np \text{ and } \mathrm{Var}(X) = np(1-p).$$
3. Let $X \sim$ Geometric(p) with $p \in (0, 1]$ then
$$E[X] = \frac{1}{p} \text{ and } \mathrm{Var}(X) = \frac{1-p}{p^2}.$$
4. Let X take $1, 2, 3, \ldots$ values with $P(X = k) = \frac{6}{\pi^2}\frac{1}{k^2}$ for all $k \in \{1, 2, 3, \ldots\}$ then
$$E[X] \text{ is not defined; and thus } \mathrm{Var}(X) \text{ is not defined.}$$

Variance of a Continuous Random Variable

Examples:

1. Let $X \sim$ Uniform$[a, b]$ with $a < b$, that is,

$$f_X(x) = \begin{cases} 0, & \text{if } x < a, \\ \frac{1}{b-a}, & \text{if } a \leq x \leq b, \\ 0, & \text{if } x > b, \end{cases}$$

then

$$E[X] = \frac{a+b}{2} \quad \text{and} \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

2. Let $X \sim N(0, 1^2)$, i.e. be *standard normal*, that is,

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for all } x \in \mathbb{R},$$

then

$$E[X] = 0 \quad \text{and} \quad \mathrm{Var}(X) = 1.$$

Contents

Covariance and Correlation

Def.

Let X and Y be two random variables. Then, provided the expectations exist,

- *Covariance* of X and Y is defined as

$$\mathrm{Cov}(X) = \mathrm{E}\left[(X - \mathrm{E}[X])\,(Y - \mathrm{E}[Y])\right].$$

- *Correlation* of X and Y is defined as

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)Var(Y)}},$$

  provided covariance exists and $\mathrm{Var}(X), Var(Y) \neq 0$.

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\text{Cov}(X, Y) = \text{E}[XY] - E[X]E[Y]$

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - E[X]E[Y]$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$

Properties of Covariance

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2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - E[X]E[Y]$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
4. $\mathrm{Cov}(X, a) = 0$

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - E[X]E[Y]$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
4. $\mathrm{Cov}(X, a) = 0$
5. $\mathrm{Cov}(X, Y) = 0$ if X and Y are independent

Properties of Covariance

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2. $\text{Cov}(X, X) = \text{Var}(X)$
3. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
4. $\text{Cov}(X, a) = 0$
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6. $\text{Cov}(aX, bY) = ab\,\text{Cov}(X, Y)$

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - E[X]E[Y]$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
4. $\mathrm{Cov}(X, a) = 0$
5. $\mathrm{Cov}(X, Y) = 0$ if X and Y are independent
6. $\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y)$
7. $\mathrm{Cov}(X, Y + Z) = \mathrm{Cov}(X, Y) + \mathrm{Cov}(X, Z)$

Properties of Covariance

Let X, Y, Z, and W be random variables and $a, b, c \in \mathbb{R}$ be some constants. Assuming covariances exist,

1. $\mathrm{Cov}(X, Y) = \mathrm{E}[XY] - E[X]E[Y]$
2. $\mathrm{Cov}(X, X) = \mathrm{Var}(X)$
3. $\mathrm{Cov}(X, Y) = \mathrm{Cov}(Y, X)$
4. $\mathrm{Cov}(X, a) = 0$
5. $\mathrm{Cov}(X, Y) = 0$ if X and Y are independent
6. $\mathrm{Cov}(aX, bY) = ab\,\mathrm{Cov}(X, Y)$
7. $\mathrm{Cov}(X, Y + Z) = \mathrm{Cov}(X, Y) + \mathrm{Cov}(X, Z)$
8. $\mathrm{Cov}(aX + bY, cZ + dW) =$
   $ac\,\mathrm{Cov}(X, Z) + ad\,\mathrm{Cov}(X, W) + bc\,\mathrm{Cov}(Y, Z) + bd\,\mathrm{Cov}(Y, W)$

Properties of Covariance

THEOREM A

Suppose that $U = a + \sum_{i=1}^{n} b_i X_i$ and $V = c + \sum_{j=1}^{m} d_j Y_j$. Then

$$\mathrm{Cov}(U, V) = \sum_{i=1}^{n} \sum_{j=1}^{m} b_i d_j \mathrm{Cov}(X_i, Y_j) \qquad \blacksquare$$

COROLLARY A

$\mathrm{Var}(a + \sum_{i=1}^{n} b_i X_i) = \sum_{i=1}^{n} \sum_{j=1}^{n} b_i b_j \mathrm{Cov}(X_i, X_j)$. $\blacksquare$

COROLLARY B

$\mathrm{Var}(\sum_{i=1}^{n} X_i) = \sum_{i=1}^{n} \mathrm{Var}(X_i)$, if the X_i are independent. $\blacksquare$

Source: *Mathematical Statistics and Data Analysis* by John Rice

Properties of Covariance

Thm.
Let X and Y be two random variables. Then

$$-1 \le \rho_{X,Y} \le 1,$$

provided the correlation exists. Furthermore,

$$\rho_{X,Y} = \pm 1 \text{ if and only if } P(Y = a + bX) = 1$$

for some constants a and b.